

MUHAMMAD LABIYB AFAKH

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SUMMARY

Robotics Engineer and PhD Candidate with 7+ years developing autonomous robot systems using C, C++, and Python. Specialized in system integration, computer vision for robotics, real-time control and navigation algorithms. Experience deploying ROS-based and native solutions for manipulation and mobile platforms with proven real-time performance.

- Experience in autonomous navigation systems, collision avoidance, and path tracking across aerial, mobile, water surface, and manipulator robotic platforms.
- Full-stack robotics development capability: building complete autonomous systems from scratch including mechanical design and manufacturing, electrical hardware development, embedded or low-level programming, system integration, and high-level planning.
- Strong embedded systems background with extensive hardware debugging experience including serial communication protocols, socket programming and sensor interface troubleshooting.
- Achieved 200Hz real-time system performance through optimized communication and multi-sensor data processing.

PROFESSIONAL EXPERIENCES

Computer Vision and Robot Motion

October 2025-now

FingerVision, Inc. (Intern)

- Integrating existing object detection models (YOLO) with FingerVision tactile sensors for grasp object classification and placement pose estimation.
- Implementing ROS-based motion control, integrating vision perception with MoveIt trajectory planning for UR3 manipulation.
- Collecting and annotating dataset for fine-tuning vision models.
- Working on imitation learning from data collection, training, and validation.

Robotics Engineer

April 2024 - March 2025

QibiTech Inc. (Part-Time)

- Developed ROS2 packages for HR4C manipulator with MoveIt integration, implementing hardware interfaces supporting position, velocity, and effort control modes, along with action servers and test suite.
- Developed robot description (URDF), visualization, and simulation on pybullet/gazebo.
- Improved the system integration between Kachaka AMR and HATS.
- Investigated the manipulability and navigation of a mobile manipulator robot using Gazebo and Robotics Toolbox Python simulation.

Research and Development Engineer

Jan 2022 - March 2024

Shanghai Micro Ears Ingeniousness Intelligent Technology Co., Ltd. (Part-Time)

- Designed and manufactured the robot structure.
- Developed embedded control software and sensor data processing pipeline.
- Optimized serial communication protocol and data packet structure, achieving 200Hz sampling rate with real-time GUI visualization.
- Developed and integrated system with a manipulator robot (myCobot 320).

Freelance Robotics and Embedded System Engineer

2018-2019

- Built an embedded system for solar tracker to move the solar panel by given position that sent from base-station.
- Built a small-device for weighing system used in a gas company.
- Built embedded system for pull and press machine including monitoring of the pressure.

SKILLS

Programming	C++ (primary), Python, C
Robotic Systems	ROS1/ROS2, MoveIt, ROS Navigation, SLAM, Path Tracking & Navigation, Sensor Fusion, Obstacle Avoidance
Real-time Control	Embedded Systems, Serial Communication, Control Systems, PID/Fuzzy Controllers
Computer Vision	OpenCV, PyTorch, Object Detection and Instance Segmentation (YOLO and MaskRCNN), Visual Servoing, Camera Calibration
Simulation	Gazebo, MuJoCo, PyBullet, V-REP
Development Tools	Git, CI/CD, Docker, Linux
Design & Fabrication	CAD (Fusion 360, SolidWorks), CAM, PCB Design (KiCad, EasyEDA, Eagle)
Hardware	ARM-based (STM32, NXP/Teensy), ESP32, ATmega, IMU, 2D and 3D Lidar, GNSS, Camera
Robot Platforms	UR3, Kachaka Pro, HR4C, myCobot 320, Kinova Jaco, Seed-Noid-Mover
Languages	Indonesian (Native), English (Fluent), Japanese (Beginner)

EDUCATION

Doctor of Mechanical Systems Engineering , Tokyo Metropolitan University	Mar 2026 (Expected)
Master of Mechanical Systems Engineering , Tokyo Metropolitan University	Oct 2019 - Sept 2021
• Relevant Courses: Robotic System Design, Intelligent Robot, Ubiquitous Robotics	
Bachelor of Computer Engineering , Electronics Engineering Polytechnic Institute of Surabaya	2014 - 2018
• Relevant Courses: Embedded System, Robot System & Control, Computer Vision, Intelligent System, Robotic and Automation, and Real-Time Operating System	

HONOR AND MEMBERSHIP

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| • Institute of Electrical and Electronics Engineers (IEEE) Society | (current) |
| • IEEE Robotics and Automation Society Membership | (current) |
| • The Japan Society of Mechanical Engineers | (current) |
| • Second Runner-up Customer Interaction Task Category, World Robot Summit FCSC @IFAC | 2023 |
| • Runner Up Toilet Cleaning Robot Category, World Robot Summit FSCS Trial Competition | 2019 |
| • Second Runner Up of ABU Robocon 2016 | 2016 |
| • Winner ABU Robocon Indonesia Contest | 2016 |

ACADEMIC EXPERIENCES

Research Assistant (Part-Time) Tokyo Metropolitan University	Jan 2025 - Present <i>Japan</i>
• Developing unmanned water surface vehicle with autonomous navigation, focusing on system integration, debugging, and ROS2 migration	
• Developing 6-DoF lightweight manipulator including actuator using 3D-printed materials.	

- Continuing flapping-wing aerial vehicle research and system optimization

Tokyo Metropolitan University Arena Reframing (AR) Research Fellow
Tokyo Metropolitan University

Apr 2023 - Sept 2024
Japan

- Conducted research on flapping-wing aerial vehicle development and performance optimization.

Research Assistant (Part-Time)
Tokyo Metropolitan University

Oct 2021 - Mar 2023
Japan

- Developed mechanical structure of servo-driven flapping-wing robot system and parameters observation.

Student Intern
BRIN-Aviation and Space Research Organization

Jan 2017 - Feb 2017
Indonesia

PUBLICATIONS

Journals

- **Agile Maneuvering of Servo-Driven Flapping Robot: Novel Approach Using Center Flapping Angle and Offset Angle Variables.** IEEE/ASME Transactions on Mechatronics. (Co-author)
- **Performance Evaluation and Wing Deformation Analysis of Flapping-Wing Aerial Vehicles with Varying Flapping Parameters and Patterns,** Journal of Robotics and Mechatronics, Vol.36, No.5, 2024/10.
- **Senswing: A Force Sensing Wing for Intelligent Flapping-Wing Aerial Vehicles—Wing Design and Comprehensive Evaluation of Force Sensing Capabilities.,** Journal of Robotics and Mechatronics, Vol.36, No.5, 2024/10. (Co-author)
- **Study Towards a Flapping Robot Maintaining Attitude During Gliding.** International Journal on Advance Science, Engineering, Information, and Technology (IJASEIT) · Jan 22, 2023.

Conferences

- **Servo-Driven Flapping Robot That Uses Its Tail for Self-Standing Takeoff.** 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025). (Co-author)
- **Servo-Driven Flapping-Wing Aerial Vehicle (FWAV) Payload Capacity and Navigation Performance.**The 2025 IEEE/SICE International Symposium on System Integration (SII).
- **Development of Flapping Robot with Self-Takeoff from The Ground Capability.** IEEE International Conference on Robotics and Automation (ICRA) · Oct 18, 2021
- **Bicycle Path Planning on Omnidirectional Mobile Robot Using Fuzzy Logic Controller.** IEEE · Aug 7, 2018
- **Implementation of PID Controller in Active Ball Handling System of Middle Size Robot Soccer.** IEEE · Aug 7, 2018. (Co-author)
- **Aksara Jawa Text Detection in Scene Images using Convolutional Neural Network.** IEEE · Sep 27, 2017

PROJECTS

Six-leg Robot Locomotion. January 2025 - Present
Contribution:

- Integrate existing RL algorithms in MuJoCo simulation environment.

Ornibibot, an agile flapping micro aerial vehicle (FMAVs). Oct 2019 - Present.

This project develops an ornithopter that can be a helpful robot for future in the society. This project is a part of my academic path from master student to PhD student (current).

Contribution:

- Developed several robot's mechanical structure including design and manufacture.
- Developed robot's hardware utilizing several kinds of micro-controller (ARM based and ESP32) and single board computer (ARM based).
- Developed robot's low-level and high-level system and controller with or without ROS1/ROS2.
- Implemented PID controller with given Inertial Measurement data to perform attitude control using tail mechanism.
- Improved the performance serial communication by optimizing the packet data so it could work in above 500Hz.
- Conducted comprehensive system analysis by integrating force sensors and motion capture cameras to measure wing deformation and thrust generation.
- Improved system capabilities through mechanical optimization achieving 100g payload capacity, while implementing on board collision avoidance utilizing 8x8 ToF and 50Hz UDP communication with base station.
- Implemented outdoor localization by using GNSS and IMU sensor and tested VSLAM using ORB-SLAM3 or existing VIO algorithms.
- Enhancing collision avoidance performance by upgrading from 8x8 ToF sensor to PMD Flexx2 depth or point-cloud and validating the collision avoidance in MuJoCo simulation environment.

Skills: C, C++ (ROS1/ROS2), Python, CAD, CAM, EDA, MuJoCo, Computer Vision

Source Code: [OrnibiBot](#) — [OrnibiBotMicro](#) — [ObstacleAvoidance](#) — [CollisionAvoidance](#) — [OrnibiBotESP](#)

Unmanned Water Surface Vehicle (January - March 2025)

Developed autonomous water surface vehicle with advanced navigation and control systems, focusing on ROS2 migration and comprehensive system architecture design. Key Achievements:

- Migrated legacy ROS1 codebase to ROS2 for improved real-time performance and distributed system capabilities
- Debugged and optimized code and system for reliable autonomous operation (navigation and SLAM).

Skills: C++, ROS2

Autonomous Service Robot with SLAM (May - July 2023)

Led development of autonomous service robot for retail environment using manipulator, SLAM and computer vision.

Key Achievements:

- Led programming team for World Robot Summit competition (2nd runner-up).
- Integrated multiple subsystems: manipulator pick and place, LiDAR SLAM, Kinect vision, and navigation stack.
- Configured and tuned nav2 autonomous navigation stack with dynamic obstacle avoidance for retail environment.
- Integrated and improved object recognition pipeline using fine-tuned YOLO models for grasp pose estimation.

Skills: Python, ROS, MoveIt, PyTorch, SLAM, Computer Vision

Repository: [WRS2023](#)

Autonomous Mobile Robot Navigation System - Bachelor Thesis (2017-2018)

Developed advanced navigation system for omnidirectional robot soccer platform with autonomous path recovery capabilities after collision or obstacle interference.

Project Overview: This bachelor thesis project addressed the challenge of navigation recovery when robots collide with obstacles or other robots during autonomous operation. The system was validated through V-REP simulation environment by implementing bicycle path tracking algorithms.

Key Achievements:

- Implemented bicycle path tracking algorithm with fuzzy logic controller that dynamically adjusts look-ahead distance and robot velocity, significantly reducing overshoot during path recovery
- Developed real-time PID controllers for precise individual motor speed control and robot orientation
- Integrated wheel odometry and IMU data for accurate position estimation and localization
- Successfully demonstrated smooth path recovery after external force disturbances, with robot automatically returning to desired trajectory
- Designed and implemented ROS-based control architecture with 3 STM32 microcontrollers and 1 Arduino Mega microcontrollers(3 for motor control, 1 for odometry) each interfacing directly with ROS high-level planning nodes via serial communication.
- Validated system performance through simulation testing where external forces were applied to knock robot off-path, demonstrating autonomous recovery capabilities

Skills: C/C++, STM32, Embedded Systems, ROS, PID/Fuzzy Controllers, Qt, V-REP Simulation

Repository: [FuzzyBicyclePathTracking](#)

Unmanned Fast Boat PENShip. Oct 2016 - Dec 2016.

Built autonomous water surface vehicles with visual servoing navigation system, successfully demonstrating path-following capabilities in competitive environment.

- Implemented visual servoing control for centerline tracking using custom color-based detection in OpenCV, successfully competing with both catamaran and monohull platforms in USV Contest 2016.
- Led team coordination and system integration efforts

Skills: C/C++, Computer Vision (OpenCV), Embedded Systems

ABU Robocon 2016

Key Achievements:

- Developed embedded control systems for coordinated two-robot autonomous operation
- Implemented navigation algorithms using odometry feedback for precise robot positioning and guidance
- Integrated multiple feedback sensors (IMU, encoders, proximity sensors) for robust system operation
- Achieved significant performance improvements through iterative optimization: 36 seconds (regional), 23 seconds (national), and 18 seconds (international competition)
- Successfully demonstrated non-contact robot guidance, autonomous object manipulation, and climbing capabilities
- Led system integration, testing, and performance optimization achieving 70% improvement over previous competition results
- Won ABU Robocon Indonesia Contest 2016 and achieved 2nd Runner-up in international competition

Skills: C, STM32, Embedded Systems, Autonomous Navigation, Sensor Integration